

VISION-BASED ADAPTIVE DRIVE ASSIST SYSTEM FOR ELECTRIC BIKES

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INTRODUCTION

Urban mobility in developing economies such as Sri Lanka faces mounting pressure as fossil fuel prices rise and traffic congestion worsens. Electric bikes (e-bikes) have emerged as a practical micro-mobility solution; however, as a class of Vulnerable Road User (VRU), e-bike riders account for a disproportionate share of road traffic fatalities. The World Health Organization estimates that VRUs represent more than 50% of the 1.35 million annual road deaths worldwide. Traditional e-bike speed control relies on open-loop throttle hardware and rudimentary pedal-assist sensors that provide no contextual awareness of the surrounding traffic environment, creating a persistent safety gap.

Advanced Driver Assistance Systems (ADAS) offer a proven pathway to improve road safety, yet their adaptation to lightweight two-wheeled platforms remains largely unexplored. This research presents a Vision-Based Adaptive Drive Assist System (V-ADAS) designed specifically for electric bikes. The system fuses a monocular camera with ultrasonic sensors, deploying lightweight deep-learning models YOLOv26n for real-time vehicle detection and TwinLiteNet for lane segmentation on a Raspberry Pi 5 edge computing platform. A closed-loop PID controller, custom electromechanical braking actuator, and haptic lane-departure warning complete the integrated safety stack.

METHOD

The development followed a structured iterative methodology encompassing requirements analysis, model selection, hardware design, software integration, and empirical validation.

A custom dataset was assembled by mounting a 5 MP Raspberry Pi Camera on an e-bike and capturing 20 minutes of video on Sri Lankan urban roads. Ten vehicle categories were annotated using Roboflow: bike, tuk_tuk, van, mini_van, truck, mini_truck, SUV, bus, sedan car, and person. Bounding-box labels were used for vehicle detection; instance-segmentation polygon labels for lane lines. The dataset was split 70% training 20% validation 10% testing.

Five candidate architectures were benchmarked on the Raspberry Pi 5. YOLOv8n, YOLOv11n, YOLOv26n, a TensorFlow Lite INT8 model, and FastestDet-V2. Evaluation criteria were inference latency (ms/frame), throughput (FPS), accuracy (mAP50-95), model size, and RAM usage. YOLOv26n compiled to the NCNN inference framework achieved the best balance, delivering 9.71 FPS and 40.7% mAP50-95 at only 9.3 MB.

Four approaches were compared for lane detection model MobileNetV3, TwinLiteNet (INT8/ONNX), OpenCV colour thresholding, and Fast-SCNN. TwinLiteNet was selected for its 12.6 FPS throughput at only 1.74 MB, providing more than three times the frame rate of MobileNetV3 while maintaining an acceptable 35.5% mIoU sufficient for reliable lane-departure detection in real time.

The central processing unit is a Raspberry Pi 5 (8 GB). Low-level actuation is managed by dual ESP32 microcontrollers: an ESP32-S3 for PID speed control and stepper-motor braking, and an ESP32-C3 for wheel-speed telemetry via hall-effect pulses. Automatic braking is achieved through a NEMA 17 stepper motor coupled to a repurposed two-stage worm-and-helical gearbox (55:1 ratio), producing a maximum output torque of 1 540 Ncm. The combined braking system (CBS) routes dual steel cables to front and rear brake arms simultaneously. Two mini vibration motors on the handlebars deliver left right haptic alerts for lane departure.

The PID speed controller was physically tuned on a dynamic bench. A deadband offset (PWM = 70) was applied to overcome static friction, and a rate-limiter was added to prevent aggressive throttle transients. The optimised parameters $K_p = 6.5$, $K_i = 0.56$, $K_d = 3.0$ produced smooth acceleration with minimal steady-state error and no overshoot oscillations.

Longitudinal distance is computed using the pinhole camera model: $d = (H \times f) / h_{pix}$, where H is the class-specific real-world vehicle height, f is the calibrated focal length, and h_{pix} is the detected bounding-box pixel height. A first-order exponential moving average (EMA) smoothing filter suppresses vibration-induced noise in the distance signal. The ByteTrack algorithm maintains temporal identity across frames.

Table 1 Key ADAS Features Implemented in the V-ADAS System

ADAS Feature	Function	E-bike Relevance
Forward Collision Warning (FCW)	Detects imminent collisions using camera and radar, alerts the rider	Prevents rear-end collisions at urban speeds
Automatic Emergency Braking (AEB)	Automatically applies brakes when collision is imminent	Reduces crash severity for e-bikes
Object/Obstacle Detection	Identifies pedestrians, vehicles, and obstacles in real-time	Core safety function for e-bike ADAS
Lane Departure Warning (LDW)	Monitors lateral position and warns of unintended lane deviation	Prevents side-by-side collisions in urban lanes

FINDINGS

The YOLOv26n model, trained for 224 epochs on the custom Sri Lankan urban dataset, achieved a mean Average Precision (mAP@0.5) of 0.758 across all ten vehicle classes. High-confidence classes included tuk_tuk (F1 = 0.812, mAP = 0.888), bike (F1 = 0.864, mAP = 0.810), and bus (F1 = 0.813, mAP = 0.836). Moderate performance on SUV (mAP = 0.611) and mini_van (mAP = 0.723) reflects limited training samples for those classes. The normalized confusion matrix confirmed high true-positive rates for safety-critical targets with minimal cross-class confusion.

Table 2 Object Detection Model Benchmark on Raspberry Pi 5

Model	FPS (Avg)	Accuracy (mAP50-95)	Size	RAM Usage
YOLOv8n	9.57	37.1	12 MB	973 MB
YOLOv11n	9.08	39.3	10.1 MB	1073 MB
YOLOv26n (Selected)	9.71	40.7	9.3 MB	1002 MB
TensorFlow INT8	3.53	36.5	40.2 MB	1296 MB
FastestDet-V2	11.53	23.0	18 MB	1146 MB

TwinLiteNet (INT8/ONNX) achieved 12.6 FPS with a 35.5% mIoU on the test set more than three times faster than MobileNetV3 (4 FPS, 45.8% mIoU). The OpenCV colour-thresholding baseline attained 30 FPS but collapsed to 21.3% mIoU under shadow and glare, making it unsuitable for outdoor operation. Fast-SCNN produced a near-zero 1.4% mIoU, indicating incompatible pre-trained weights on this dataset. TwinLiteNet was therefore selected as the best overall solution, balancing real-time responsiveness with acceptable accuracy for Lane Departure Warning triggering.

Table 3 Lane Detection Model Benchmark on Raspberry Pi 5

Model	FPS	mIoU (%)	Size	CPU Usage
MobileNetV3	4	45.8	12.5 MB	62%
TwinLiteNet INT8 (Selected)	12.6	35.5	1.74 MB	83%
OpenCV Color Detection	30	21.3	4.3 kB	15%
Fast-SCNN	4	1.4	236 kB	78%

Static validation across 63 samples per vehicle class showed an overall maximum absolute distance error of 1.1 m (observed for buses and pedestrians), with average errors near zero for tuk_tuk (0.042 m), car (0.030 m), and mini_truck (0.058 m). Dynamic validation under active riding conditions reduced the maximum error to 0.8 m, with average errors below 0.04 m for most classes, confirming robust performance during motion.

Static bench tests across 100 data points mapped braking lag distance against initial speed. At 4–6 km/h the average stopping lag was 0.47 m; at 26–28 km/h it reached 9.89 m. The Minimum Safe Distance (MSD) formulation incorporating system latency ($t_{sys} = 0.3$ s), maximum deceleration ($\max = 1.0$ m/s²), sensor error margin (1.1 m), surface friction variance, and mechanical wear yielded an MSD of 4.75 m at 7 km/h. A practical extended set-point of 7 m is applied at 8 km/h, scaling to 20 m at 20 km/h, providing real-world safety margins beyond theoretical minimums.

The tuned PID controller ($K_p = 6.5$, $K_i = 0.56$, $K_d = 3.0$) achieved stable speed tracking with a dampened response, smooth acceleration profile, and a small steady-state error. Safety interlocks successfully triggered motor cutoff and stepper-motor braking whenever vehicle speed exceeded the set-point by more than 5 km/h in bench testing.

CONCLUSION

This research successfully developed and validated a Vision-Based Adaptive Drive Assist System for electric bikes, demonstrating that real-time perception, adaptive cruise control, and automatic emergency braking can be achieved on a low-cost Raspberry Pi 5 edge platform. The YOLOv26n model (NCNN-compiled) and TwinLiteNet (INT8/ONNX) ran concurrently without computational bottlenecks, delivering 9.71 FPS object detection and 12.6 FPS lane segmentation respectively. The custom 55:1 gearbox-coupled stepper-motor braking system produced maximum output torques of 1 540 Ncm and validated stopping distances consistent with the theoretically derived MSD model. Distance estimation errors remained within 1.1 m under static conditions and 0.8 m under dynamic conditions adequate for reliable ACC set-point control.

The system addresses a documented research gap, no prior work has presented a fully integrated, real-world-deployed, vision-based ADAS for electric bikes that simultaneously handles obstacle detection, adaptive speed regulation, automatic braking actuation, and lane departure warning on a single edge device. The design is specifically optimized for narrow, partially marked roads typical of South Asian urban environments.

Future work will focus on expanding the training dataset to improve detection of under-represented classes (SUV, mini_van), integrating nighttime and adverse-weather resilience, performing field trials with human riders to quantify cognitive workload reduction, and conducting a formal cost-benefit analysis for commercial retrofit deployment.

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